These types of limitations in the classical definition of probability led to the evolution of the modern definition of probability which is based on the concept of sets. It is known an axiomatic approach.

The foundations of the Modern Probability theory were laid by Andrey Nikolayevich Kolmogorov, a Russian mathematician who combined the notion of sample space introduced by Richard von Mises, and measure theory and presented his axiomatic system for probability theory in 1933. We introduce the axiomatic approach proposed by A.N. Kolmogorov. Based on this, it is possible to construct a logically perfect structure of the modern theory of probability theory. The classical theory of probability is a particular case of axiomatic probability. The axioms are a set of rules, which can be used to prove theorems of probability.



A.N. Kolmogorov

12.4.2 Axiomatic approach to Probability

Axioms of probability

Let S be a finite sample space, let $\mathcal{P}(S)$ be the class of events, and let P be a real valued function defined on $\mathcal{P}(S)$. Then $P(A)$ is called probability function of the event A, when the following axioms are hold:

$[P_1]$	For any event A,	$P(A) \geq 0$	(Non-negativity axiom)
$[P_2]$	For any two mutually exclusive events		
		$P(A \cup B) = P(A) + P(B)$	(Additivity axiom)
$[P_3]$	For the certain event	$P(S) = 1$	(Normalization axiom)

Note 12.1

(i) $0 \leq P(A) \leq 1$

(ii) If $A_1, A_2, A_3, ..., A_n$ are mutually exclusive events in a sample space S, then

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_n) = P(A_1) + P(A_2) + P(A_3) + \cdots + P(A_n)$$

Theorems on finite probability spaces (without proof)

When the outcomes are equally likely Theorem 12.1 is applicable, else Theorem 12.2 is applicable.

Theorem 12.1

Let S be a sample space and for any subset A of S, let $P(A) = \frac{n(A)}{n(S)}$. Then $P(A)$ satisfies axioms of probability $[P_1], [P_2]$, and $[P_3]$.

Theorem12.2

Let S be a finite sample space say $S = \{a_1, a_2, a_3, ..., a_n\}$. A finite probability space is obtained by assigning to each point a_i in S a real number p_i, is called the probability of a_i, satisfying the following properties:

(i) Each $p_i \geq 0$. (ii) The sum of the p_i is 1, that is, $\sum p_i = p_1 + p_2 + p_3 + \cdots + p_1 = 1$.

If the probability $P(A)$, of an event A is defined as the sum of the probabilities of the points in A, then the function $P(A)$ satisfies the axioms of probability $[P_1], [P_2]$, and $[P_3]$.

Note: Sometimes the points in a finite sample space and their assigned probabilities are given in the form of a table as follows:

Outcome	a_1	a_2	a_3	...	a_n
Probability	p_1	p_2	p_3	...	p_n

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